

PRAKTIKUM BASIS DATA

MODUL 11
NoSQL_Key_Value_Database



Oleh:

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TP

1. A.

```
Cassandra CQL Shell
Connected to Test Cluster at 127.0.0.1:9042.
[cqlsh 5.0.1 | Cassandra 2.2.3 | CQL spec 3.3.1 | Native protocol v4]
Use HELP for help.
WARNING: pyreadline dependency missing.  Install to enable tab completion.
cqlsh> CREATE KEYSPACE spotifyne_0802
... WITH replication = {
...     'class' : 'SimpleStrategy',
...     'replication_factor' : 1
... };
cqlsh> USE spotifyne_0802;
cqlsh:spotifyne_0802> |
```

B.

```
Cassandra CQL Shell
Connected to Test Cluster at 127.0.0.1:9042.
[cqlsh 5.0.1 | Cassandra 2.2.3 | CQL spec 3.3.1 | Native protocol v4]
Use HELP for help.
WARNING: pyreadline dependency missing.  Install to enable tab completion.
cqlsh> CREATE KEYSPACE spotifyne_0802
... WITH replication = {
...     'class' : 'SimpleStrategy',
...     'replication_factor' : 1
... };
cqlsh> USE spotifyne_0802;
cqlsh:spotifyne_0802> CREATE TABLE musician_data (
...     artist_id text PRIMARY KEY,
...     artist_name text,
...     followers int,
...     origin text,
...     genres set<text>
... );
cqlsh:spotifyne_0802> |
```

```
Cassandra CQL Shell
cqlsh> USE spotify_0802;
cqlsh:spotify_0802> CREATE TABLE musician_data (
...     artist_id text PRIMARY KEY,
...     artist_name text,
...     followers int,
...     origin text,
...     genres set<text>
... );
cqlsh:spotify_0802> INSERT INTO musician_data (
...     artist_id,
...     artist_name,
...     followers,
...     origin,
...     genres
... )
... VALUES (
...     'AR001',
...     'The Weeknd',
...     95000000,
...     'Canada',
...     {'Pop', 'R&B'}
... );
cqlsh:spotify_0802> INSERT INTO musician_data (
...     artist_id,
...     artist_name,
...     followers,
...     origin,
...     genres
... )
... VALUES (
...     'AR002',
...     'Bruno Mars',
...     78000000,
...     'USA',
...     {'Pop', 'Funk'}
... );
cqlsh:spotify_0802> INSERT INTO musician_data (
...     artist_id,
...     artist_name,
...     followers,
...     origin,
...     genres
... )
... VALUES (
...     'AR002',
...     'Bruno Mars',
...     78000000,
...     'USA',
...     {'Pop', 'Funk'}
... );
cqlsh:spotify_0802>
```

```
Cassandra CQL Shell
... );
cqlsh:spotify_0802> INSERT INTO musician_data (
...     artist_id,
...     artist_name,
...     followers,
...     origin,
...     genres
... )
... VALUES (
...     'AR002',
...     'Bruno Mars',
...     78000000,
...     'USA',
...     {'Pop', 'Funk'}
... );
cqlsh:spotify_0802> INSERT INTO musician_data (
...     artist_id,
...     artist_name,
...     followers,
...     origin,
...     genres
... )
... VALUES (
...     'AR002',
...     'Bruno Mars',
...     78000000,
...     'USA',
...     {'Pop', 'Funk'}
... );
cqlsh:spotify_0802> SELECT * FROM musician_data;
-----
artist_id | artist_name | followers | genres | origin
-----
AR001    | The Weeknd | 95000000 | {'Pop', 'R&B'} | Canada
AR002    | Bruno Mars | 78000000 | {'Funk', 'Pop'} | USA
(2 rows)
cqlsh:spotify_0802>
```

2. Keuntungan Menggunakan tipe data set

Keuntungan utama menggunakan tipe data set pada kolom genres adalah karena satu musisi dapat memiliki lebih dari satu genre tanpa perlu membuat tabel tambahan seperti pada database relasional. Dengan tipe data set, data genre dapat disimpan langsung dalam satu kolom secara lebih sederhana dan efisien. Selain itu, tipe data set juga mencegah adanya data genre yang duplikat serta mempermudah proses pengambilan data pada aplikasi dengan jumlah pengguna dan data yang sangat besar seperti Spotify.